

Department of Computer Science and Engineering (V Semester)			
JAVA and Spring Framework Lab (CS3603-1) AY-2026-27			
<u>Mini-Project Abstract</u>			
Project Title: AI Cricket Analytics & Fantasy Team Advisor			
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Abstract (Max 500 words) Cricket fans and fantasy-league players often struggle to extract meaningful insights from the vast volume of statistics generated across T20, ODI and Test cricket, as existing platforms either overwhelm users with raw scorecards or offer generic, non-personalized advice. This project proposes an AI-powered cricket analytics platform that allows users to ask natural-language questions about real player and team statistics (e.g., “Which batter has the highest strike rate in the death overs over the last 10 T20 matches?” or “How do left-arm spinners perform at Chepauk?”) and receive accurate, data-grounded answers along with fantasy-team recommendations. The system periodically fetches scorecard and ball-by-ball data from public cricket data sources through a scheduled sync job and stores it in a relational database. When a user submits a question through the chat interface, the backend sends the question along with the database schema to a Large Language Model (LLM), which translates it into a structured SQL query (text-to-SQL). The generated query is validated so that only read-only SELECT statements are executed, and the results are passed back to the LLM to generate a natural-language explanation or scouting-style summary. A parallel fantasy-advisor module uses the same data pipeline to suggest a balanced playing XI along with captain and vice-captain picks, based on recent player form, head-to-head records, venue history and pitch conditions. Key objectives include integrating regularly updated cricket data with an LLM for natural-language querying, translating user questions into safe, structured database queries with human-readable responses, and providing fantasy-team recommendations based on player form, role balance (batters, bowlers, all-rounders and wicket-keepers) and match-up analysis. The expected outcome is a working web application that lets users explore cricket statistics conversationally and receive AI-generated insights and fantasy recommendations, demonstrating the practical integration of enterprise backend engineering with modern AI reasoning patterns such as text-to-SQL and LLM-based summarization.			
Tools and Technologies Used: Backend: Spring Boot (REST APIs, scheduled tasks), Spring Data JPA/Hibernate Database: PostgreSQL (relational schema for players, teams, matches, venues, innings, deliveries) Data Source: Cricsheet (open ball-by-ball data) and CricketData.org API (fixtures and live scorecards) via scheduled sync job AI Layer: LLM (GPT-4o-mini / Llama 3) for text-to-SQL generation and natural-language reasoning Frontend: React (chat interface, stats tables, charts via Chart.js)			
Approved: Yes / No		Date:	Staff Name and Signature:

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